* Theoretical Analyses and Modelling: 1 K hybrid Cryocooler: 4stage Stirling Pulse tube + JTC

↳ Objective will be to determine operation objectives based design parameters:

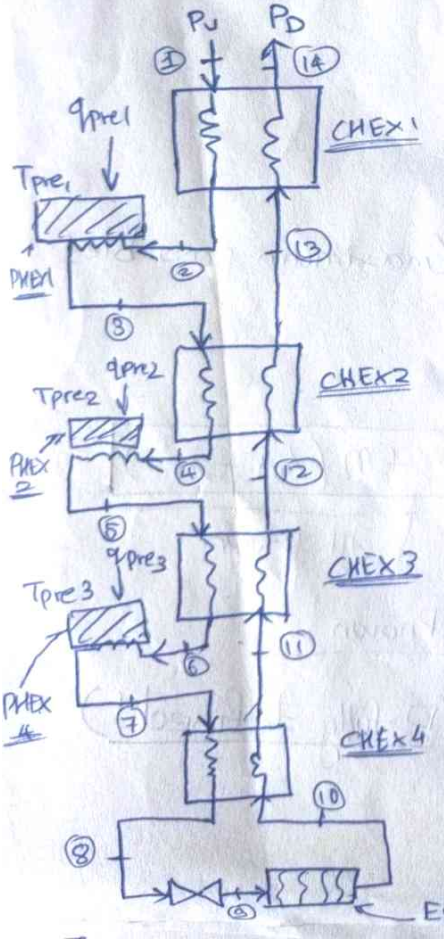
- T_c = Cold end of JT (here 1K)
- P_u = Upstream pressure of JT (compressor dependent) ← To be set.
- Q_c = Heat intake capacity (load) of the end at T_c

↳ Cooler consists of 3 precoolers at 2nd, 3rd, 4th stage of Stirling Cryocooler. Say those are maintained at T_i, and need to intake Q_{prei} heat.

∴ Objective will be to find P_u, T_i, Q_{prei} such that the Stirling Pulse tube & JTC could be fully design, just based on T_c & Q_c required.

↳ The design also includes 4 Counter flow HX (CHEX). Say their efficiencies $\eta_1, \eta_2, \eta_3, \eta_4$ are also known / or requirement for a decent Q_c output could be posed

↳ Consider the schematic for JTC:



* Hard assumptions:

1] No pressure loss befor & after JT valve

↳ ∴ State 1, 2, 3, 4, 5, 6, 7, 8 at P_u ← upstream pressure.

↳ State 9, 10, 11, 12, 13, 14 at P_D ← Downstream pressure.

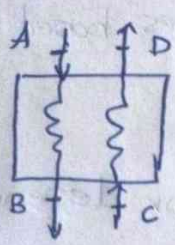
2] Temperature after PHEX = Temp of PHEX

∴ $T_3 = T_{pre1}$, $T_5 = T_{pre2}$

$T_7 = T_{pre3}$

3] Working fluid is Helium-3.

↳ Consider a CHEX, say it effectiveness is η.



↳ Say line AB has pressure P_u
Line CD has pressure P_d

↳ Now say we know T_A & T_C.

↓
Why, cause T_A mostly $= T_{pre}$
and $T_C = T_D$ of the previous CHEX

↓
Form coupled equations which can be solved implicitly.

∴ For this HX:

Consider a fictitious state: $\underline{\underline{A'}}$ which is at T_A, P_d.

↳ Now, we know: $A: (T_A, P_u, h_A)$, $C: (T_c, P_d, h_c)$, $A': (T_A, P_d, h_{A'})$ — A, C, A' fully defined!

$h_A = h(P_u, T_A)$
$h_C = h(P_d, T_C)$
$h_{A'} = h(P_d, T_A)$

↳ By the definition of ~~actual~~ effectiveness,

$$\eta = \frac{Q_{actual}}{Q_{max}}$$

For the downstream: $Q_{Max} = h_{A'} - h_c$ (maximum possible)

$$Q_{actual} = h_D - h_c$$

$$\therefore \eta = \frac{h_D - h_c}{h_{A'} - h_c} \rightarrow \boxed{h_D = h_c + \eta(h_{A'} - h_c)}$$

All known

$\therefore$ h_D known!

↳ h_D, $P_D = P_d$ $\Rightarrow$ T_D could be found. ($\square$ fully defined!)

↳ Energy balance! Say mass flows are equal;

$$\dot{m}(h_A - h_B) = \dot{m}(h_D - h_C)$$

$$\therefore h_A - h_B = h_D - h_C$$

$$\therefore h_B = -h_D + h_C + h_A$$

$$\therefore \boxed{h_B = h_A - \eta(h_A' - h_C)}$$

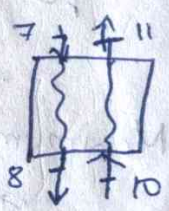
All known.

$\therefore h_B, P_B = P_u \Rightarrow T_B$ could be known! $\therefore$ B fully defined!

↳ Let's simplify the analysis and say $\eta \to 1$ for all CHEX.

Then; consider CHEX 4:

For the temperatures we are operating at; we see:



Upstream, $C_{Pu} < C_{PD}$

For upstream $\therefore T_{Pu} \to T_{Pd}$ $\therefore T_8 \to T_{10}$ → Cold-end pinch

Illy for downstream, $C_{Pu} > C_{PD}$

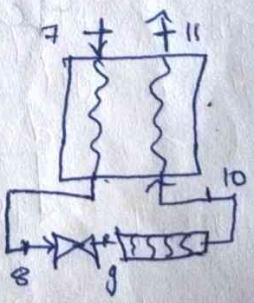
$\therefore T_{11} \to T_7$ → Hot-end pinch

* JT cold end analysis :-

↳ We said $\eta_4 = 1$ $\therefore \boxed{q_c = h_{10} - h_9}$

↳ 8 → 9 is JT throttling. ⇒ Isenthalpic

$\therefore \boxed{h_8 = h_9}$.



$\therefore q_c = h_{10} - h_8$

↳ Also, For CHEX, $h_{11} - h_{10} = h_7 - h_8$

$\therefore q_c = h_{10} - h_7 = h_{10} - h_8$

$\eta_4 = \frac{h_{11} - h_{10}}{h_{7D} - h_{10}}$ $\therefore$ $\boxed{q_c = h_{10} - h_7 + \eta_4 (h_{7D} - h_{10})}$

$\downarrow \eta_4 = 1$

$\therefore$ $\boxed{q_c = h_{7D} - h_7}$

↳ Let's define $\boxed{\Delta h_T = h_{7D} - h_7}$

↳ Now physically, after throttling, state 9 will have some liquid at T_c, say quality is x_g. Suppose at stage 10, we have a quality x_{10}. Now $x_{10} \leq 1$ (it will still be at T_c since that temperature is to be maintained).

$\therefore$ $q_c = h_{10} - h_g = (x_{10} - x_g) h_{fg}|_{T_c}$

↳ Maximum possible liquid will form when $x_{10} = 1$

$\therefore$ $\boxed{q_L = (1 - x_g) h_{fg}|_{T_c}}$

↳ We had earlier said $q_c = \Delta h_T$

$\therefore$ $q_c = \Delta h_T = (x_{10} - x_g) h_{fg}|_{T_c}$

$= (x_{10} - 1) h_{fg}|_{T_c} + \underbrace{(1 - x_g) h_{fg}|_{T_c}}_{q_L}$

$\therefore$ $\Delta h_T = q_L + \underbrace{(x_{10} - 1)}_{\leq 0} h_{fg}|_{T_c}$

$\therefore$ $\boxed{q_L \geq \Delta h_T}$

↳ But say Δh_T was higher than q_L. That would mean $x_{10} > 1$ → Not possible.

$\therefore q_c > \Delta h_T = (x_{10} - x_g) h_{fg}|_{T_c}$ not realisable.
($x_{10} > 1$)

↳ It is basically showing the tendency to superheat.

∴ In that case we limit our cooling to latent heat of maximum possible heating, which is q_L

$\therefore$ If $q_L \geq \Delta h_T$, $q_c = \Delta h_T$

$q_L < \Delta h_T$, $q_c = q_L$

$$\boxed{q_c = \min [\Delta h_T, q_L]}$$

↳ q_c depends on Δh_T & q_L

↳ $\Delta h_T = h_{7D} - h_7$ → P_d depends on T_c cause $P_d = P_{sat}|_{T_c}$

↳ $\therefore \Delta h_T = f(P_0, T_{pres}, T_c)$
(T_7)

↳ $q_L = (1 - x_g) h_{fg}|_{T_c}$

↳ Depends on T_c

$x_g = \dfrac{h_g - h_f|_{T_c}}{h_{fg}|_{T_c}} = \dfrac{h_8 - h_f|_{T_c}}{h_{fg}|_{T_c}}$ ($\because T_{sat} = h_g$)

& h_8 depends on CHEX; which for $\eta_4 = 1$ depends on T_{pres} & T_c

$\therefore q_L = f(P_0, T_{pres}, T_c)$

$\therefore q_c = f(P_u, T_{pre_3}, T_c)$

↳ Now, Let's fix T_{pre_3} & T_c (=1k) and see variation of q_c with P_u.

↳ For operating range, $P_u \uparrow \Rightarrow \eta_{th} \uparrow, q_L \downarrow$

opposing trends → Maxima produced. (Fig 5)

↳ Let's say the pressure where q_c maximum occured as $P_{u,opt}$ and q_c as $q_{c,opt}$.

$\therefore$ For a set T_c &, each T_{pre_3} has $P_{u,opt}$ & q_{copt}.

Illy, conversely, for a set T_c, each P_u has a $T_{pre_3,opt}$ & q_{copt}

$\therefore$ We have triplet pairs of (T_{pre}, P_u, q_c)

3D contour

↳ Let's say for various T_{pre_3} we obtain $P_{u,opt}$ & fit a equation (curve) $\therefore P_{u,opt} = f(T_{pre3})$

(To avoid calculation)

Illy, $q_{opt} = f(T_{pre_3}, P_{uopt})$

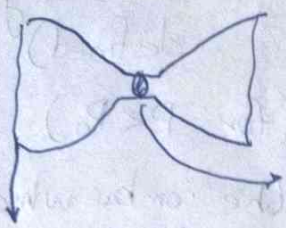
$= f(T_{pre_3})$

↳ On the 3D contour we have a maxima.

Maxima of $q_{c,opt} = f(T_{pre3})$

↳ It occurs at $8k = T_{pre_3}$, $P_u = 0.36$ MPa & $P_d = 1.16$ kPa ($T_c = 1k$)

↳ Let's consider JT valve as a CD nozzle:



Throat: $P^*, T^*, \rho^*, V^*, h^*, s^*$

Stagnation (inlet)
P_0, T_0, h_0, s_0

↳ Consider velocity → 0 here

↳ For a CD nozzle: Mass flux for an ideal gas flow is:

$$\left[G_{ID}(T_0, P_0) = \gamma^{1/2} \left(\frac{2}{1+\gamma} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \sqrt{\frac{M P_0^2}{R_u T_0}} \right]$$

↳ Here, $\gamma = c_p / c_v$ (for He 3 ideal gas, $\gamma = 5/3$)

M = Molar mass of He 3

R_u = Univ gas const

↳ But at operating conditions, He 3 is strongly real in behaviour.

∴ We define actual Mass flux: $G = \Gamma \cdot G_{ID}$

Γ ↓ correction factor.

↳ To find G & Γ:

$$h(p,T) + \frac{1}{2} V^2 = h(P_0, T_0)$$ (anywhere in the nozzle)

← Energy balance.

↳ For throat:

$$\left[h(p^*, T^*) + \frac{1}{2} (V^*)^2 = h(p_0, T_0) \right]$$

→ But at throat, $V^* = a|_{T^*}$, a is speed of sound at T^* ($M=1$)

Also, CD nozzle is ~~isent~~ isentropic:

$$\therefore s(p^*, T^*) = s(p_0, T_0).$$

$$\therefore \underline{s^* = s_0}$$

↳ Now (P_0, T_0) is (P_8, T_8). ∴ State '8' governs inlet.

↳ We say that (P_0, T_0) is known (inlet, or state 8 is fully defined)

∴ • Find (P, T, h) combos which have $s = s_0$ (for $P < P_0$)

• Now for all those, $\left[h(p,T) + \frac{a^2}{2} = h_0\right]$, the combo which satisfy this will give us the required throat condition.

↳ Now, once throat condition (state) is fully defined (by P, T, h):

$$G = \text{Mass flux} = \frac{\dot{m}}{A} = \rho^* V^*$$

$$\therefore G = \rho^* V^* = \rho^* a^* \quad \text{(known for the fully defined state of throat)}$$

∴ G through nozzle is known!

∴ $\Gamma = G / G_{ID}$ could be found for (P_0, T_0).

↳ Let us find / fit a curve such that $\Gamma = f(P_u, T_{pres})$.

$$\therefore G = f(P_u, T_{pres})$$

↳ Now, $\dot{m} = \frac{\pi d^2}{4} \times G = \frac{\pi d^2}{4} \times \Gamma \times G_{ID}$

$$\therefore Q_c = \dot{m} \cdot q_c = \frac{\pi d^2}{4} \times \underbrace{\Gamma \times G_{ID}}_{f(T_{pres}, P_u)} \times q_c$$

↳ Now, an overall optimum Q_{cmax} could be found for (T_{pres}, P_u)